Network Description

This project is about designing a small network using Cisco Packet Tracer. The network contains three routers (Internet, Active, and Standby), two switches, and two PCs. The main goal is to provide communication between all devices and keep the network working even if one router or link fails.

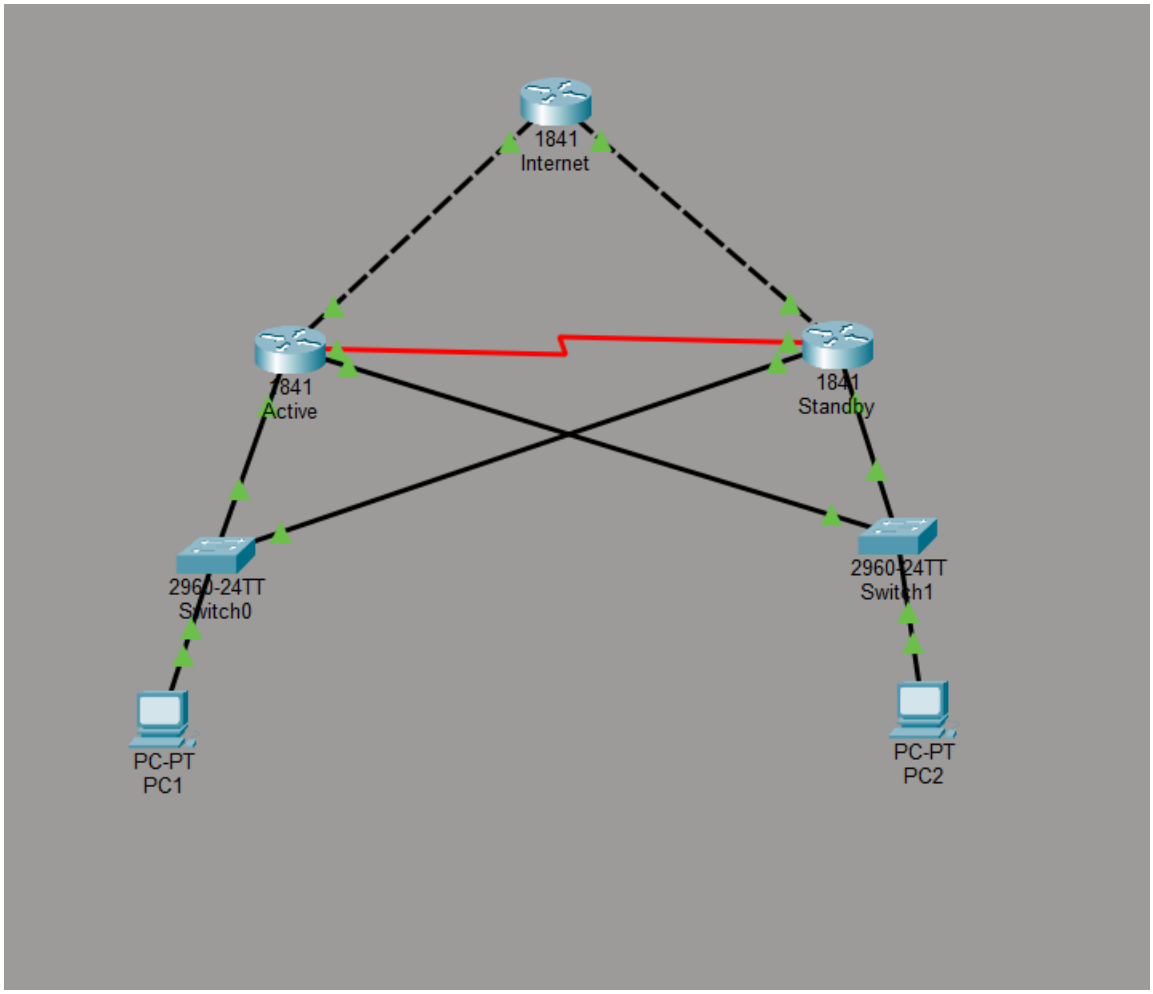
The network uses OSPF as the routing protocol to exchange routing information between routers and automatically find the best path for sending data. All routers are configured in Area 23.

The project also uses HSRP Version 2 to provide gateway redundancy. The Active router forwards traffic during normal operation, while the Standby router is ready to take over if the Active router fails. This helps reduce network downtime and keeps users connected.

For security, a warning banner and passwords are configured on the routers and switches to prevent unauthorized access.

Overall, this network demonstrates the use of OSPF, HSRP, IP addressing, and basic network security to build a reliable and stable enterprise networ

Network Topology Design/



Network Connectivity Test (Ping Results)

pc1/

```
C:\>ping 30.30.30.2

Pinging 30.30.30.2 with 32 bytes of data:

Reply from 30.30.30.2: bytes=32 time<1ms TTL=254
Request timed out.
Reply from 30.30.30.2: bytes=32 time<1ms TTL=254
Reply from 30.30.30.2: bytes=32 time=2ms TTL=254

Ping statistics for 30.30.30.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 20.20.20.50

Pinging 20.20.20.50 with 32 bytes of data:

Request timed out.
Reply from 20.20.20.50: bytes=32 time=1ms TTL=127
Reply from 20.20.20.50: bytes=32 time=2ms TTL=127
Reply from 20.20.20.50: bytes=32 time<1ms TTL=127

Ping statistics for 20.20.20.50:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms
```

OSPF Routing Table – Active Router/

```
Active#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    20.0.0.0/8 is directly connected, Ethernet0/0/0
C    30.0.0.0/8 is directly connected, FastEthernet0/1
O    40.0.0.0/8 [110/2] via 10.10.10.3, 00:07:42, FastEthernet0/0
                [110/2] via 20.20.20.2, 00:07:42, Ethernet0/0/0
                [110/2] via 30.30.30.2, 00:07:42, FastEthernet0/1
C    50.0.0.0/8 is directly connected, Serial0/1/0
```

OSPF Routing Table – Standby Router/

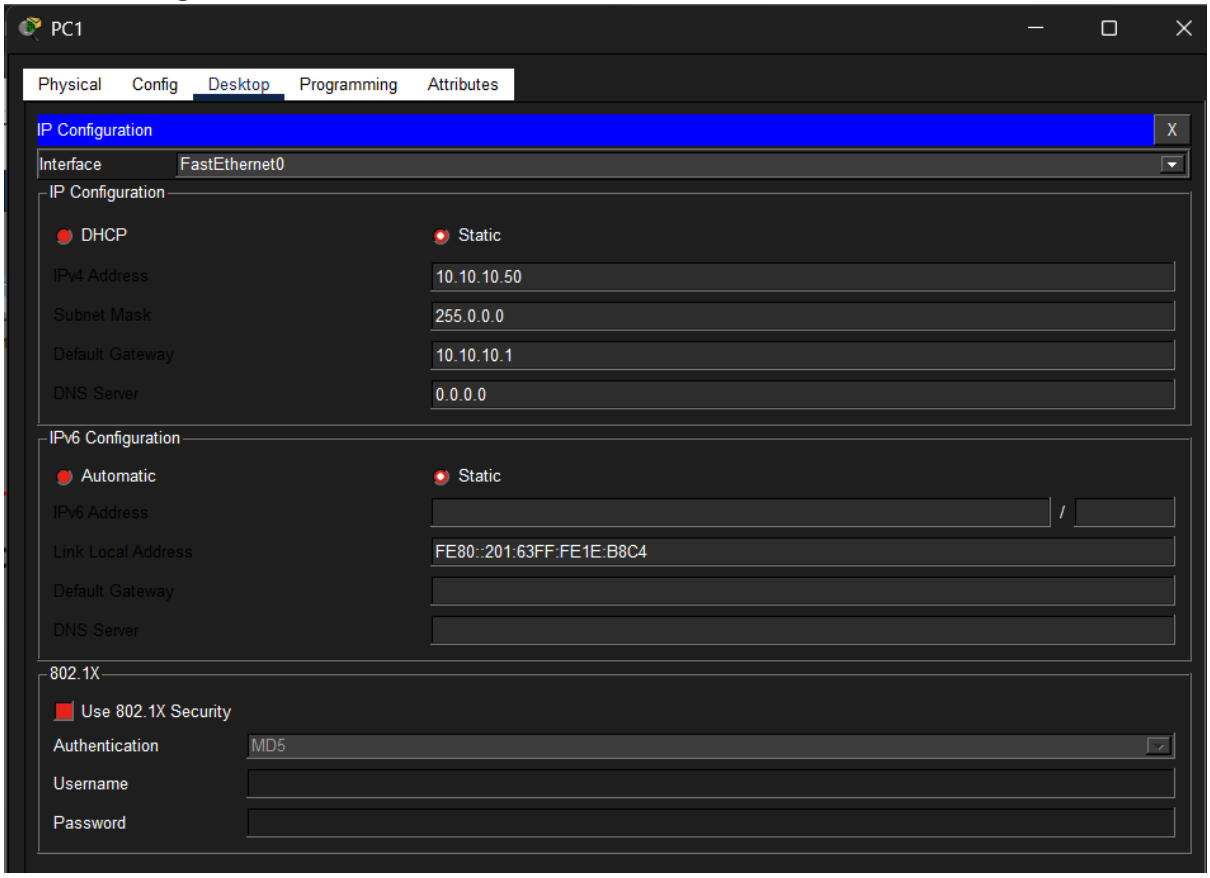
```
Standby#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, Ethernet0/0/0
C    20.0.0.0/8 is directly connected, FastEthernet0/0
O    30.0.0.0/8 [110/2] via 10.10.10.2, 00:09:06, Ethernet0/0/0
                [110/2] via 20.20.20.3, 00:09:06, FastEthernet0/0
                [110/2] via 40.40.40.2, 00:09:06, FastEthernet0/1
C    40.0.0.0/8 is directly connected, FastEthernet0/1
C    50.0.0.0/8 is directly connected, Serial0/1/0

Standby#
```

PC1 IP Configuration/



HSRP Failover Test – PC1/

```
C:\>tracert 20.20.20.50

Tracing route to 20.20.20.50 over a maximum of 30 hops:

  1    0 ms      1 ms      1 ms      10.10.10.3
  2    *         2 ms      1 ms      20.20.20.50

Trace complete.
```

HSRP Recovery Test – PC1/

```
C:\>tracert 20.20.20.50

Tracing route to 20.20.20.50 over a maximum of 30 hops:

  1    0 ms      1 ms      0 ms      10.10.10.2
  2    0 ms      0 ms      0 ms      20.20.20.50

Trace complete.
```

HSRP Failover Test – PC2/

```
C:\>tracert 30.30.30.1

Tracing route to 30.30.30.1 over a maximum of 30 hops:

  1    0 ms      1 ms      0 ms      30.30.30.1

Trace complete.
```

HSRP Recovery Test – PC2/

```
C:\>tracert 30.30.30.1

Tracing route to 30.30.30.1 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      30.30.30.1

Trace complete.

C:\>
```

HSRP Configuration (Active and Standby Routers)/

```
Active#show standby
FastEthernet0/0 - Group 10 (version 2)
  State is Active
    13 state changes, last state change 11:19:27
  Virtual IP address is 10.10.10.1
  Active virtual MAC address is 0000.0C9F.F00A
    Local virtual MAC address is 0000.0C9F.F00A (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.834 secs
  Preemption enabled
  Active router is local
  Standby router is 10.10.10.3, priority 99 (expires in 7 sec)
  Priority 111 (configured 111)
  Group name is hsrp-Fa0/0-10 (default)
Ethernet0/0/0 - Group 20 (version 2)
  State is Active
    8 state changes, last state change 00:00:39
  Virtual IP address is 20.20.20.1
  Active virtual MAC address is 0000.0C9F.F014
    Local virtual MAC address is 0000.0C9F.F014 (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 2.49 secs
  Preemption enabled
  Active router is local
  Standby router is 20.20.20.2, priority 99 (expires in 7 sec)
  Priority 111 (configured 111)
  Group name is hsrp-Eth0/0/0-20 (default)
Active#
```

```
Standby#show standby
FastEthernet0/0 - Group 20 (version 2)
  State is Standby
    35 state changes, last state change 11:22:10
  Virtual IP address is 20.20.20.1
  Active virtual MAC address is 0000.0C9F.F014
    Local virtual MAC address is 0000.0C9F.F014 (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.85 secs
  Preemption disabled
  Active router is 20.20.20.3, priority 111 (expires in 8 sec)
    MAC address is 0000.0C9F.F014
  Standby router is local
  Priority 99 (configured 99)
  Group name is hsrp-Fa0/0-20 (default)
Ethernet0/0/0 - Group 10 (version 2)
  State is Standby
    27 state changes, last state change 11:19:57
  Virtual IP address is 10.10.10.1
  Active virtual MAC address is 0000.0C9F.F00A
    Local virtual MAC address is 0000.0C9F.F00A (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.6 secs
  Preemption disabled
  Active router is 10.10.10.2, priority 111 (expires in 7 sec)
    MAC address is 0000.0C9F.F00A
  Standby router is local
  Priority 99 (configured 99)
  Group name is hsrp-Eth0/0/0-10 (default)
```

OSPF, Banner, and Password Configuration/

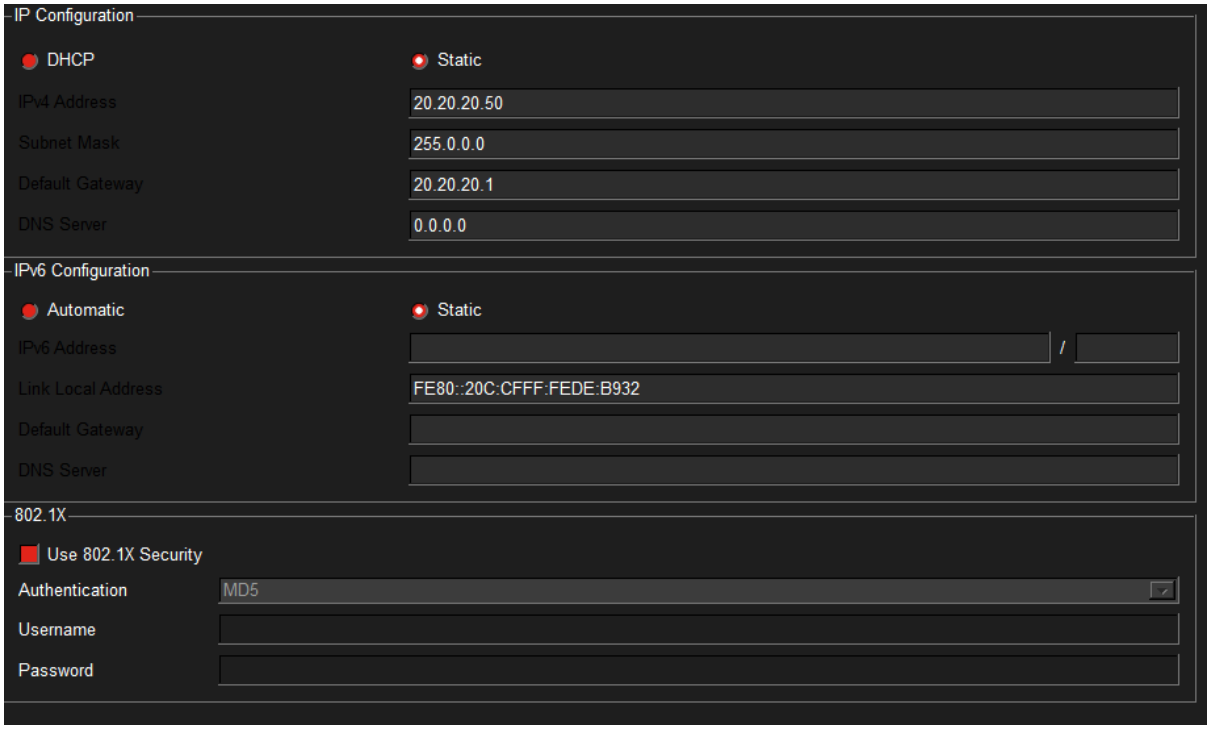
Active# show running-config
Building configuration...

Current configuration : 1351 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Active
!
!
!
enable secret 5 1mERr$zZYfVZ9RllN6hRVDE983h0
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!

```
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface FastEthernet0/0
ip address 10.10.10.2 255.0.0.0
duplex auto
speed auto
standby version 2
standby 10 ip 10.10.10.1
standby 10 priority 111
standby 10 preempt
!
interface FastEthernet0/1
ip address 30.30.30.1 255.0.0.0
duplex auto
speed auto
!
interface Ethernet0/0/0
ip address 20.20.20.3 255.0.0.0
duplex auto
speed auto
standby version 2
standby 20 ip 20.20.20.1
standby 20 priority 111
standby 20 preempt
!
interface Serial0/1/0
ip address 50.50.50.1 255.0.0.0
clock rate 64000
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
```

```
no ip address
shutdown
!
router ospf 1
router-id 1.1.1.1
log-adjacency-changes
network 10.0.0.0 0.255.255.255 area 23
network 20.0.0.0 0.255.255.255 area 23
network 30.0.0.0 0.255.255.255 area 23
network 50.0.0.0 0.255.255.255 area 23
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^CFor Authorized IT Staff Only^C
!
!
!
!
!
line con 0
password summer473admin
login
!
line aux 0
!
line vty 0 4
password summer473admin
login
!
!
!
end
```

PC2 IP Configuration/


IP Configuration
DHCP
Static
IPv4 Address
20.20.20.50
Subnet Mask
255.0.0.0
Default Gateway
20.20.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20C:CFFF:FEDE:B932
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password